Rapid Hyperspectral Anomaly Detection Using Discriminative Band Selection

Hao-Fang Yan, Yong-Qiang Zhao, *Member, IEEE*, Jonathan Cheung-Wai Chan, and Seong G. Kong, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) exhibits high-quality spectral signals that convey subtle differences, enabling the discrimination of similar materials and providing a unique advantage for anomaly detection (AD). Fine spectral of anomalies can be effectively identified amidst heterogeneous background pixels. Given the similarity of materials in spatial and spectral dimensions, joint utilization of spatial and spectral information enhances detection performance. However, many existing AD approaches for HSIs usually achieve high accuracy at the expense of high computational complexity. In response to the requirements of practical detection scenarios–efficiency, robustness, and accuracy–this article introduces a rapid and robust AD algorithm through discriminative band selection for HSIs. We propose a spatial–spectral feature extraction strategy to ensure detection accuracy. Initially, to effectively mine context information across a broad spectral range, the HSI cube in space is partitioned into several groups using a coarse-to-fine strategy. Subsequently, we identify the most relevant and informative bands based on spatial local density and spectral information entropy, forming the coarse HSI bands subset. Following this, we design a multiband target–background ratio (MBTBR) to capture strongly discriminative bands, resulting in the fine HSI bands subset. Finally, we present an adaptively spatial—spectral feature extraction strategy to detect anomalous targets. Extensive experimental results on real hyperspectral datasets demonstrate that the proposed method achieves satisfactory performance compared to the state-of-the-art algorithms, validating its strong robustness and low computational complexity simultaneously.**

***Index Terms*— Adaptive fusion, coarse-to-fine band selection strategy, discriminative band, hyperspectral anomaly detection (HAD), spatial—spectral feature extraction.**

I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data cubes [1]. Each pixel in an HSI is considered a high-dimensional vector, providing richer characteristic information about target objects [2], [3], [27]. Handling hundreds of dimensional data remains a daunting task. The large number of bands in HSI increases the complexity of data processing, and the high correlation between bands leads to problems of high redundant information and high computational complexity, making it challenging to meet real-time requirements in practical applications. Hyperspectral anomaly detection (HAD) relies on spectral and spatial anomaly characteristics to detect anomalous targets without requiring prior information about the target and background, attracting significant attention worldwide [6], [29], [61], [62], [63]. Regarding spectral anomaly characteristics, the spectrum of an anomalous target in an HSI distinctly contrasts with its surrounding background [4], [5], [7]. In terms of spatial anomaly characteristics, anomalous targets are often small and exhibit a sparse distribution in HSIs [8]. Current HAD algorithms primarily rely on spectral differences between abnormal targets and the background, ignoring spatial characteristics. Therefore, reducing the interference of redundant data to improve the efficiency of data processing and ensuring efficient spatial–spectral feature extraction are two key issues faced by HAD in practice.

Band selection aims to choose a subset of bands with low correlation from the original hyperspectral data, without losing physical meaning and information, while preserving the inherent characteristics of hyperspectral data. Common band selection methods are based on ranking, combining, and clustering. Ranking-based methods usually utilize metrics such as information entropy [9] to evaluate the importance of each band and select the top K-bands as a subset. The selected bands often have a high correlation. Combining-based methods select band subsets by maximizing band information and minimizing band correlation, as seen in methods like optimal exponential factor (OIF) [10]. Exhaustive subset selection severely limits the calculation speed. Clustering-based methods divide the bands into multiple groups with low correlation between the groups and then select representative spectra from each group, for instance, the adaptive subspace partition strategy (ASPS) [11]. These methods neglect the effect of selected band subset, which only evaluates the single band. Nevertheless, these methods either choose the most informative bands or cluster centers to form a subset, making it challenging for the selected subset to be both discriminative and informative simultaneously.

In the past few decades, researchers have developed numerous HAD methods. The well-known Reed-Xiaoli detector (RXD) [12] assumes that the spectral characteristics of HSI

Manuscript received 1 March 2024; revised 14 June 2024 and 24 July 2024; accepted 23 August 2024. Date of publication 29 August 2024; date of current version 14 October 2024. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 61771391, in part by the Key Research and Development Plan of Shaanxi Province under Grant 2020ZDLGY07-11, and in part by the Fundamental Research Funds for the Central Universities, Science, Technology and Innovation Commission of Shenzhen Municipality under Grant JCYJ20170815162956949 and Grant JCYJ20180306171146740. *(Corresponding author: Yong-Qiang Zhao.)*

Hao-Fang Yan and Yong-Qiang Zhao are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: yanhaofang@mail.nwpu.edu.cn; zhaoyq@nwpu.edu.cn).

Jonathan Cheung-Wai Chan is with the Department of Electronics and Informatics, Vrije Universiteit Brussel, 1050 Brussels, Belgium (e-mail: jonathan.chan@VUB.be).

Seong G. Kong is with the Department of Computer Engineering, Sejong University, Seoul 05006, South Korea (e-mail: skong@sejong.edu).

Digital Object Identifier 10.1109/TGRS.2024.3451559

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

background follow a Gaussian distribution, which is too strict to be satisfied in reality. To circumvent the inherent deficiency of the inappropriate data distribution model in Reed-Xiaoli (RX), many other algorithms have also been proposed [15], [16], [17]. Scholars also employed representation models to solve the HAD problem. The selection-based discriminative forest (SSDF) algorithm exploited the spatial anomaly characteristics of HSI [18], exhibiting low complexity but poor robustness. Based on low rank and sparse matrix decomposition, Ma et al. [19] proposed an integration of feature extraction and background purification anomaly detector (FEBPAD). Assuming that the background occurs in lowest rank representation of the image, Xu et al. [20] proposed a low-rank and sparse representation (LRASR) algorithm, separating anomalies from the background. Huyan et al. [21] proposed PAB-DC, assuming a low-rank background and sparse anomaly, to decompose HSIs into three types: background, anomaly, and noise. Hou et al. [22] proposed a spectral–spatial fusion HAD (SSFAD) method that performs saliency enhancement on spectral features. Gao et al. [58] propose a blind-spot network (BS3LNet) trained in self-supervised strategy, in which the training patch pairs of blind spots are generated from a single HSI. Nevertheless, most HAD algorithms fail to fully utilize the spectral and spatial characteristics of anomaly targets, leading to a deterioration in their detection accuracy and robustness, some algorithms require the construction of online dictionaries, increasing their computation complexity.

HSIs exhibit both stability and similarity in the spectral domain. The spectral stability of the same substance within a certain wavelength range can affect the robust selection of bands. The spectral similarity between an anomaly target and background reduces the target–background contrast, increasing the difficulty of AD. Due to the complexity of the spectral domain, the general band selection methods mentioned above for reducing spectral redundancy are not closely associated with HAD tasks. This leads to poor effect performance when the screened band data is directly applied to HAD. Current HAD methods do not fully utilize spectral and spatial features and cannot meet real-time requirements. To address the task of rapid detection of abnormal targets in a large field of view, this article designs a semisupervised rapid hyperspectral anomaly target detection method that combines discriminative band selection with an adaptive spatial–spectral feature extraction strategy. Through a coarse-to-fine partition strategy, the band with the largest information in each cluster is selected as the candidate band. The amount of information was evaluated using spatial local intensity and spectral information entropy. Subsequently, the bands subset with the largest difference between targets and backgrounds is further screened out using the multiband target–background ratio (MBTBR). The selected bands subset serves as the input for the adaptive spatial–spectral feature extraction strategy, ultimately yielding a robust and fast AD result. A novel dual-pipeline framework is proposed for spectral and spatial feature extraction, which are then melted using the proposed adaptive fusion strategy to further highlight anomalies. This approach comprehensively considers the validity evaluation of bands and the anomaly measurement of spatial–spectral characteristics, following for the accommodation of differences in spectral characteristics among various ground objects. It overcomes spatial–spectral changes in different observation environments and effectively reflects differences between targets and backgrounds.

The main contributions of this article can be summarized as follows.

1) Introduced a "coarse-to-fine" spectral band partition method. An initial grouping strategy of an equal number of bands, and then refined it by merging the bands according to spectral similarity, resulting in groups with a different number of bands that are more dispersed and informative.
2) Leveraged spatial local density and spectral information entropy to identify key bands for AD, forming a coarse band subset. Designed an MBTBR to automatically capture strongly discriminative band within the subset, further refining it into a fine HSI band subset.
3) Proposed a dual-pipeline framework for adaptive spatial–spectral feature extraction strategy, which combines spectral features enhanced by saliency weight and spatial features enhanced by fully extracted similarity information from local patches.

The remainder of this article is arranged as follows. Section II introduces two parts: band selection and HAD. Section III presents the proposed method for HAD. In Section IV, the experimental results and related analyses are shown. Finally, Section V draws the conclusion.

II. Related Work

A. Band Selection

The high dimensionality of hyperspectral data increases the complexity of signal processing. Band selection serves as an effective method for dimensionality reduction. This involves choosing a subset that preserves the majority of structural information while discarding redundant or disruptive bands [23], [24]. Such an approach can effectively reduce the dimensionality of HSI while retaining sufficient information about ground objects, facilitating the expeditious implementation of subsequent applications.

Hyperspectral data covers a broad range of spectral bands, spanning from visible light to near-infrared and then to shortwave infrared. The initial band selection [26] is a method that evenly partitions the HSI cube based on the number of predetermined hyperspectral bands, which often serves as the baseline algorithm. Its effectiveness is limited due to the lack of a theoretical foundation. Based on wavelength coverage region, some band selection methods [30], [31] partition the hyperspectral cube according to visible light (0.4–0.7 μm), near-infrared (0.7–1.4 μm), and shortwave infrared (1.4–2.5 μm). Although these methods have a relatively reasonable theoretical basis, they fall short in considering the impact of different scenes on data acquisition, the relationship between changes in object type, and the structure of hyperspectral data. Based on neighboring transmissible correlation, Lu et al. [28] propose an automatic subspace partitioning method. The hyperspectral cube

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

is divided into multiple groups by identifying the correlation coefficient using the continuity and transferability of adjacent bands. Although this method is relatively novel, the artificially set threshold will affect the results of subsequent processing. Building on the influence of nearby bands, Wang et al. [32] employed dynamic programming to segment the HSI cube in space. Compared with the aforementioned methods, it achieves promising performance.

After dividing these bands, the next step is to select the most representative band in each group. For clustering-based methods [26], [28], [30], [31], the band nearest to the cluster center is generally chosen as the selected band. While these approaches can obtain the most relevant band, but neglect the amount of information in the band, which does not conform to the principles of band selection. In contrast to the aforementioned methods, some algorithms choose the informative band in each group by quantifying the importance of each band to obtain a subset [34], [35]. To acquire the band with the greatest difference as the most informative band, these methods analyze the difference in probability distribution in each band through non-Gaussian measurement (Kullback–Leibler (KL) divergence [14], information divergence [25], and kurtosis of probability distribution [33]). Wang et al. [32] used three algorithms, maximum-variance principal component analysis (MVPCA), enhanced fast density-peak-based clustering (E-FDPC), and information entropy, to evaluate the bands in each group. These methods either choose the most informative bands or cluster centers to form a subset, making it challenging for the selected subset to be discriminative and informative simultaneously.

Handling hundreds of dimensional data remains a daunting task, primarily due to the high correlation between bands, resulting in an exorbitant amount of redundant information and computational complexity. Therefore, a crucial issue that HAD must address in practice is how to mitigate the interference caused by redundant data and enhance the efficiency of data processing.

B. Hyperspectral Anomaly Detection

HAD has gained significant attention worldwide, as it leverages spectral and spatial anomaly characteristics to detect anomalous targets without requiring any prior information about the target and background [29]. Over the past few decades, researchers have developed numerous HAD methods, broadly categorized into three main groups: statistical model-based methods, representation model-based methods, and deep learning-based methods.

The RXD [12] is a representative method in the first category. RXD assumes that the spectral characteristics of HSI background strictly follow a Gaussian distribution. Subsequently, several modified RX algorithms were proposed [13], [14] to alleviate this strict constraint, such as weighted-RX and kernel-RX. RX-based algorithms only leverage spectral anomaly characteristics and a fixed distribution model, which may not capture all background information in HSIs, resulting in weaker robustness and lower detection accuracy. Drawing on the speed advantages of RX, Yuan et al. [60] introduce a fast HAD method via subfeatures grouping and binary accumulation (SFBA) for HSIs. This method utilizes band selection to reduce the computational complexity.

Compared to the statistical model-based methods mentioned earlier, some representation model-based methods have emerged to achieve better detection performance. Collaborative representation detector (CRD) [36] leverages the similarity relationship between nonanomalous pixels and background pixels for satisfactory detection performance. CRD linearly represents the testing pixel via its neighboring pixels, but the presence of outliers in background pixels can degrade the detection accuracy. To generate relatively pure background pixels, a CRD with outlier removal anomaly detector (CRDBORAD) [37] was proposed. Subsequently, Su et al. [38] proposed a CRD combined with principal component analysis (PCAroCRD), in which PCA was employed for outlier removal. Tan et al. [39] proposed a CRD-based local summation and inverse distance weight detector (LSADCRIDW), which makes use of the various local spatial distribution information of the neighboring pixels. Hou et al. [40] proposed an HAD method based on CRD (CRDBPSW). This method makes use of the average saliency of neighboring pixels and achieves good detection result.

Another representation model-based approach is low-rank matrix decomposition, which aims to find a low-dimensional subspace to represent the entire image with as few atoms as possible, that is, a powerful tool to exploit the intrinsic low-rank property of images [41]. Assuming that the background pixels follow a single subspace, the classical robust principal component analysis (RPCA) [41] method decomposes the HSI matrix into a low-rank matrix and a sparse matrix. Liu et al. [42] proposed the RPCA-RX anomaly detector by applying the RX detector to the sparse matrix. Low-rank representation model uses multiple subspaces to represent the data, which is suitable for an HSI with complex background features. The low-rank representation sum-to-one (LRRSTO) model [43] was proposed for AD by adding a sum-to-one constraint. Niu and Wang [44] proposed a low-rank representation based and learned dictionary (LRRaLD) method, where the dictionary is learned from the entire image with a random selection process. Tan et al. [45] proposed a single/multiple local windows-based LRRSTO (SLW/MLW_LRRSTO) method by using the local background statistics information. As of now, many low-rank representation-based methods, such as sparse representation (SRD) [46], low-rank and sparse representation (LRASR) [20], and graph total variation regularized low-rank representation (GTVLRR) [47], have been published with better detection results.

Recently, deep learning-based methods have garnered increasing attention in HAD. Li et al. [48] proposed a transferred deep convolutional neural network detector (CNND), which is trained by using pixel pairs selected from a reference dataset. With the advent of autoencoders (AEs), numerous AEs, including spectral-constrained adversarial AEs (SC_AEs) [49], stacked denoising AEs (SDAs) [50], manifold-constrained AE network [51], and a fully convolutional AE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

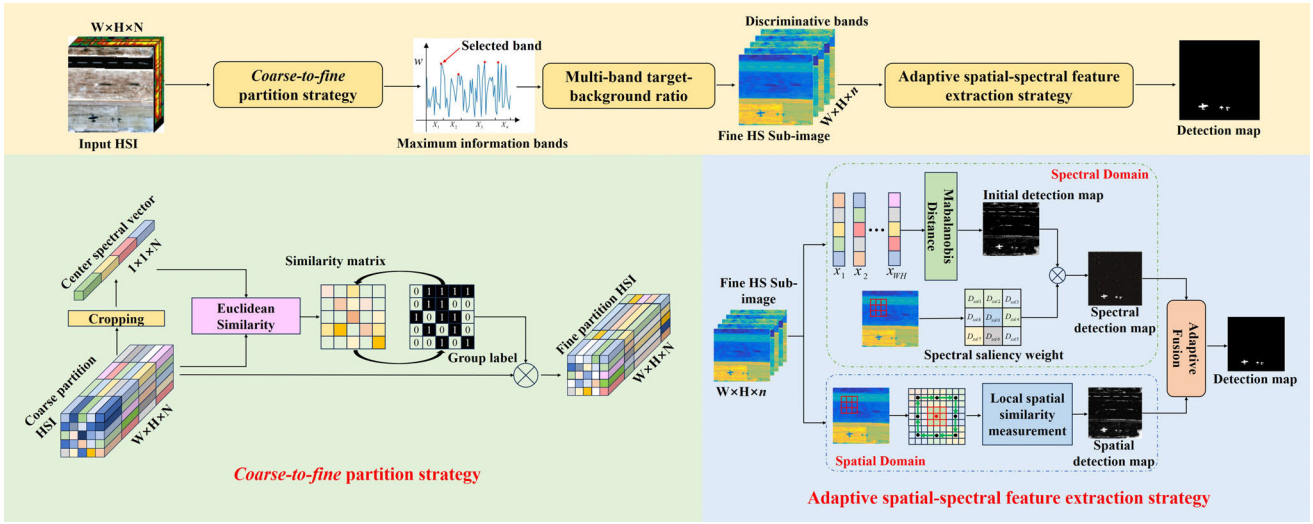


Fig. 1. Flowchart of the proposed algorithm for HAD.

with skip connections [52], were introduced into HAD. Xiang et al. [53] proposed an HAD method based on a guided AE (GAED), which reduces the feature representation for the anomaly targets. Based on Siamese network framework, Rao et al. [55], [56] solve the HAD task through similarity metric evaluation. Tu et al. [57] propose an HAD method using the spectral–spatial graph (SSG), which considers spatial and spectral information simultaneously. Tu et al. [59] treat the HAD as a frequency-domain analysis problem. The saliency-aware map obtained by phase-only reconstruction (POR) in this method significantly enhances the initial anomaly result. In order to train an efficient background reconstruction network, Wang et al. [64] proposed a pixel-shuffle downsampling blind-spot reconstruction network (PDBSNet) which operates in self-supervised mode. Wang et al. [65] presented a self-supervised blind-block network (BockNet) with a guard window. Dual-window-inspired reconstruction network (DirectNet) [66] further integrates the sliding dual-window model into a blind-block architecture.

In conclusion, the aforementioned HADs are plagued by several problems that need to be addressed: 1) traditional model-specific algorithms exhibit instability across various datasets, resulting in poor transferability; 2) most deep learning-based detectors fail to take into account spatial properties, making them susceptible to contamination from anomalies and noise; and 3) spectral similarity, along with spectral dimension, is not considered in many HAD detectors.

III. Proposed Method

This section meticulously details our proposed method. The schematic is illustrated in Fig. 1. To begin with, we analyze the challenges of spectral stability and similarity that need to be faced in practical HAD applications. Capitalizing on the characteristic of greater similarity between adjacent bands, we propose a semisupervised discriminative band selection strategy to obtain a strongly discriminative HSI bands subset. To partition the HSI cube, we introduce a coarse-to-fine grouping strategy. Within the finely partitioned groups, we calculate two factors: spatial local density and spectral information entropy. The bands exhibiting the largest weight in the product of the two factors are selected to form the coarse HSI bands subset. Subsequently, an MBTBR is employed to automatically capture the most strongly discriminative bands within the subset, culminating in the fine HSI bands subset. Finally, a novel adaptive spatial–spectral feature extraction strategy is designed for rapid HAD.

A. Spectral Stability and Similarity Analysis

HAD refers to detecting pixels with different characteristics from surrounding pixels in the scene and judging them as anomaly targets without any prior knowledge of the target. The core of AD in HSI lies in identifying potential objects that deviate significantly from the background's spectral signature. As HAD is an unsupervised process, it does not need to provide prior information about the target to be tested in advance, so it plays a great role in real life. However, three real-world challenges hinder this process. First, the spectral response value of the same substance tends to be stable in a certain wavelength range, which leads to the problems of high redundant information and high complexity. Second, low target–background contrast arises from inherent spectral similarities between the target and its surroundings, making anomalies harder to discern. Third, the spectral variability of the same substance across the scene can introduce noise and complicate robust band selection for effective detection. In this section, we analyze the spectral stability and similarity of real HSI data.

First, we extracted the spectral response values of a single substance over the full spectral range to analyze the spectral stability. The spectral response value of the anomaly target in ABU-Airport-1 data is presented in Fig. 2. For ABU-Airport-1 dataset, the three airplanes in different scales are anomalous targets, consisting of 60 pixels and accounting for 0.6% of the image. Asphalt highway, grass, and cement floor are the main background classes. As shown in Fig. 2, the spectral response values between 1030–1240 nm and 1555–1870 nm

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

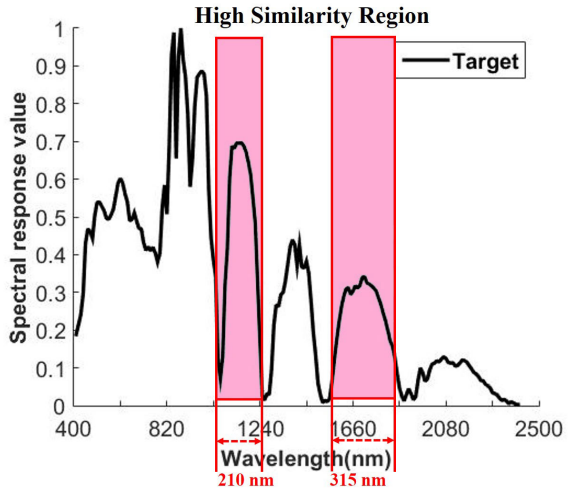


Fig. 2. Spectral stability for anomaly target in ABU-Airport-1 dataset (the wavelengths width for high similarity region correspond to 210 and 315 nm).

(pink-highlighted regions) exhibit striking similarities. These areas share the same wave type, variation trend, and amplitude. This observation aligns with a key feature of HSI data: high similarity between adjacent bands. Due to the narrow band range and repetitive wave patterns within specific regions (Fig. 2), such redundancy can contribute to the "dimensionality disaster" phenomenon, hampering subsequent AD tasks. Traditionally, addressing these challenges often involves reducing the dimensionality of the HSI cube.

After that, we analyze the spectral similarity. We studied the spectral characteristics not only of abnormal targets but also of typical ground object backgrounds (asphalt highway, grass, bridge, water, farmland, etc.) in three real HSI data. The first one is ABU-Airport-1 dataset [Fig. 3(a)]. The other two are ABU-Beach-4 dataset [Fig. 3(b)] and ABU-Urban-1 dataset [Fig. 3(c)]. For ABU-Beach-4 dataset, the vehicles on bridge are anomalous targets, consisting of 63 pixels and accounting for 0.28% of the image. Bridge, water, bare soil, and buildings are the main background classes. For ABU-Urban-1 dataset, the buildings are anomalous targets, consisting of 37 pixels and accounting for 0.37% of the image. Farmland and water are the main background classes. As shown in Fig. 3(a) and (c), the spectral variation trend and amplitude of both anomaly target and background from 820 to 1555 nm (highlighted in gray) exhibit significant overlap. Similarly, Fig. 3(b) reveals a close match in the spectral behavior of anomaly target and background from 430 to 598 nm (gray area). These bands pose a challenge for AD as they weaken the target–background contrast, making it harder to distinguish anomalies. This is because the energy received by the sensor primarily originates from the reflected light of ground objects, and factors like light intensity and distance can introduce instability in the target spectrum. Therefore, a robust band selection method should prioritize bands that exhibit high contrast between anomaly targets and backgrounds for effective AD.

To address these real-world challenges, we propose a semisupervised discriminative band selection strategy. This strategy aims to extract a subset of HSI bands that maximizes

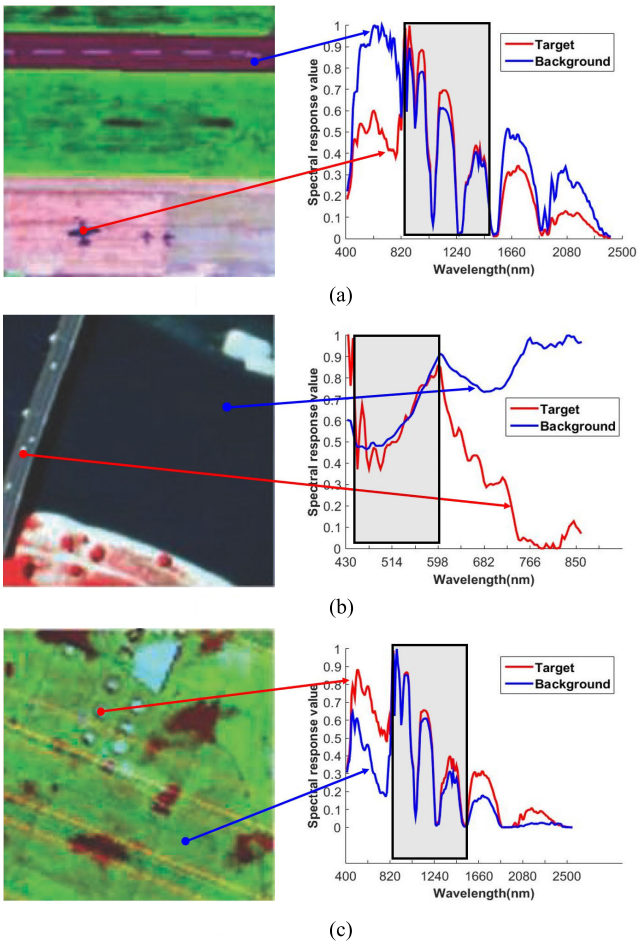


Fig. 3. Spectral comparison between the target and the background. (a) ABU-Airport-1 dataset. (b) ABU-Beach-4 dataset. (c) ABU-Urban-1 dataset.

the distinction between anomalies and the background, enabling accurate and efficient AD.

B. Semisupervised Discriminative Band Selection Strategy

This section delves into our semisupervised discriminative band selection strategy. First, we propose a coarse-to-fine partition strategy to cluster the initial HSI data. Next, within each cluster, the band exhibiting the highest combined weight of spatial local intensity and spectral information entropy is chosen to form the coarse HSI bands subset. This selection minimizes band redundancy while maintaining informative features. We propose an MBTBR to quantify the spectral distinction between anomalies and the background. This metric guides the selection of a further refined subset, focusing on bands with the largest target–background discrepancy. Finally, the resulting fine HSI bands subset is employed in the subsequent HSI AD stage, enabling fast and efficient identification of abnormal targets.

1) Coarse-to-Fine Partition Strategy: Leveraging the inherent lower similarity between nonadjacent bands, we propose a coarse-to-fine strategy for HSI cube partitioning, illustrated in Fig. 1. This strategy starts with a coarse division using neighborhood band grouping, creating initial clusters with a consistent number of bands. Subsequently, within

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

these clusters, a fine partitioning step identifies and groups bands exhibiting high similarity across specific ranges. This two-stage approach ensures a dispersed and informative subset while maintaining computational efficiency. Notably, our method focuses solely on the influence of adjacent bands, enabling faster execution while still effectively utilizing the rich contextual information embedded within the spectral data.

Let $X = \{x_1, x_2, \ldots, x_N\} \in \mathbb{R}^{WH \times N}$ be an HSI cube, where WH denotes the number of pixels per band, N is the total number of bands, and x_i is the vector of the ith band in full spectral range. The coarse partition strategy assumes that all bands are equally divided into M groups, and each group contains M bands as evenly as possible. For the original HSI cube X, they are first divided into M groups $\{X'_m\}_{m=1}^{m=M}$ in spectral dimension as evenly as possible. Concretely, every partition point G is defined as follows:

$$G(m) = \begin{cases} M, & m = 1 \\ \dfrac{(N - \mathrm{mod}(N, M)) \times m}{M}, & 2 \le m \le M - 1 \\ N, & m = M \end{cases} \quad (1)$$

where mod (:, :) is the mod operation, M is the group number, and $m = 1, 2, \ldots, M - 1$. In terms of G, $\{X'_m\}_{m=1}^{m=M}$ can be obtained by

$$X'_m = \{x_i\}_{i=G(m)}^{i=G(m+1)}, \quad m = 1, 2, \ldots, M - 1. \quad (2)$$

The result of this first partition is a coarse band grouping, which is similar to the uniform band selection method. The band number in each group is equal. Due to the lack of theoretical basis for the initial coarse division and the results are not ideal, we propose a fine neighborhood grouping approach to repartition initial groups $\{X'_m\}_{m=1}^{m=M}$, thus obtaining more accurate grouping. The band x_{λ_m} in the coarse partition groups $\{X'_m\}_{m=1}^{m=M}$ is taken as the initial cluster center, which is defined as follows:

$$\lambda_m = \left\lfloor m \times \frac{N}{M} - \frac{N}{2M} \right\rfloor, \quad m = 1, 2, \ldots, M \quad (3)$$

where m records that it currently belongs to the mth group, λ_m is the band number of initial cluster center, and $\lfloor\rfloor$ is the round down operator. To speed up the execution of the algorithm and fully dig the context information in spectral dimension, for the current cluster center x_{λ_m}, we only take into account these neighborhood bands $\{x_j\}_{j=a}^{j=b}$ around the current cluster center. We obtain a and b by (4), which is as follows:

$$(a, b) = \begin{cases} [1, \lambda_{m+1}), & m = 1 \\ (\lambda_{m-1}, \lambda_{m+1}), & 2 \le m \le M - 1 \\ (\lambda_{m-1}, N], & m = M \end{cases} \quad (4)$$

where a and b are the neighborhood boundaries of the current cluster center. We define vector R to record the similarity between the current band which wants to be repartitioned x_j and the cluster center x_{λ_m}, and initialize it to infinity. We define vector A to express the regroup label and initialize it to zero. In fine repartition strategy, the Euclidean distance $D(x_{\lambda_m}, x_j)$ between cluster center x_{λ_m} and current band which wants to be repartitioned x_j is first calculated. If $D(x_{\lambda_m}, x_j) < R(x_j)$, set $R(x_j) = D(x_{\lambda_m}, x_j)$ and $A(x_j) = m$. After vectors R and A are updated, the value in some groups is inconsistent due to the influence of factors, such as water vapor absorption or noise. For example, regroup label A is [2, 2, **1**, 2, 2, 1, 1, 1]. We process these outliers (bold label) so that each group has the same value. Through multiple iterations, the vector R and the regroup label A are updated until the regroup label A reaches a constant value. According to regroup label A, we select the maximum value of the index in each group as the new partition point to update G. The fine partition strategy of the HSI cube is conducted as $X_m = \{x_i\}_{i=G(m)}^{i=G(m+1)}$, $m = 1, 2, \ldots, M - 1$. Then the number of bands in each group is no longer consistent, the number of groups is the same as the initial. Finally, after fine partition strategy, the bands of high similarity are assigned to one group, which makes the low redundancy between different groups. Coarse-to-fine partition strategy of the HSI cube is given in Fig. 1.

2) Coarse-to-Fine Bands Subset Selection: Following the coarse-to-fine grouping of the initial HSI cube, traditional band selection methods typically focus on choosing the most relevant or informative band from each group. However, these approaches often fail to identify discriminative bands, leading to suboptimal AD performance. To address this limitation, we propose an effective coarse-to-fine bands subset selection method that chooses both informative and discriminative bands simultaneously. The core idea of coarse bands subset selection rests on ranking the bands within each group based on the product of their spatial local density and spectral information entropy. This combined weight metric ensures that only the single band with the highest discriminative potential is selected from each group. The core idea of fine bands subset selection obtains the minimum number of strongly discriminative bands.

To effectively capture the relationships between bands within each group, we leverage the K-nearest neighbors (KNNs) algorithm. This approach calculates the distances between a band and its KNNs, revealing its distribution relative to other bands within the group. For a random band x_e, the distances between it and other bands can be obtained and are arranged in ascending order according to the calculating Euclidean distance D. The KNN set of band x_e can be obtained by $\mathrm{KNN}(x_e) = \{x_u | D(x_e, x_u) \le d_{ek}\}$. x_u is a random band in the KNN set of x_e. d_{ek} is the kth distance of band x_e, where $k = 2$. To select a representative band in each group, the shared nearest neighbor (SNN) [54] is further introduced in this method to describe the number of shared neighbor elements between band x_e and band x_u, that is,

$$\begin{aligned} \mathrm{SNN}(x_e, x_u) &= |\mathrm{KNN}(x_e) \cap \mathrm{KNN}(x_u)| \\ &= |(x_u | D(x_e, x_u) \le d_{ek}) \cap (x_k | D(x_u, x_k) \le d_{uk})| \end{aligned} \quad (5)$$

where $\cap$ denotes the intersection of two bands. d_{uk} is the kth distance of band x_u, and x_k is the random band in the KNN set of x_u. SNN indicates the similarity between band x_e and K-neighbors in local space, which shows the band distribution in the space. In general, the larger this value is, the closer the distribution of band x_e and neighbor subset is. Since the number of bands is small in each group after partitioning the HSI cube, directly adopting the ratio of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

Euclidean distance and SNN of bands x_e and x_u to calculate the spatial local density could lead to statistical errors [10]. Thus, in this method, after calculating the ratio of the above two variables, we further use a Gaussian kernel function to accurately determine the spatial local density d_e

$$d_e = \sum_{x_u \in \mathrm{KNN}(x_e)} \exp\left(-\frac{D(x_e, x_u)}{\mathrm{SNN}(x_e, x_u) + 1}\right). \tag{6}$$

To ensure that the selected bands also have a large amount of spectral information, the other factor in each group is measured by calculating spectral information entropy. Information entropy [33], [34] is a statistical form of a feature, reflecting the average amount of information in the image. The larger the information entropy, the richer the spectral information. The band with the maximum spectral information entropy is selected from each cluster band as the center band of each cluster, and the spectral information entropy is used to measure the average amount of spectral information contained in the band, which is defined as follows:

$$h_e = -\sum_{WH \in \Omega} P(WH) \log P(WH) \tag{7}$$

where Ω is the grayscale color space, and $P(WH)$ is the probability of occurrence of an event in the image, obtained from the grayscale histogram.

Through the above analysis, when both factors are the largest, the selected band will not only satisfy a large amount of spatial and spectral information but also be more representative. Therefore, the weight w_e of each band in the group can be denoted as follows:

$$w_e = \begin{cases} h_e \times d_e, & \mathrm{count}(m) > k \\ h_e, & \text{otherwise} \end{cases} \tag{8}$$

where count(m) is the total number of bands in the group, and k is the parameter of KNN set. When the number of bands in the group is less than the parameter k of the nearest neighbor set, we only consider the spectral information entropy of the band because the other factor d_e has little effect. In addition, the two factors are normalized to the range of [0, 1] to keep them in the same order of magnitude by $d_e = (d_e - d_{\min})./(d_{\max} - d_{\min})$ and $h_e = (h_e - h_{\min})./(h_{\max} - h_{\min})$, in which ./ is the element division operator, $h_{\min}$ and $d_{\min}$ are the minimum of vectors h_e and d_e, and $h_{\max}$ and $d_{\max}$ are the maximum of vectors h_e and d_e. According to (8), we pick the band with largest value w_e as the most relevant and informative band in each group and thus obtain the coarse HSI bands subset $Y = \{x_1, x_2, \ldots, x_M\} \in \mathbb{R}^{WH \times M}$, which contains M bands.

In order to further reduce the computational complexity, we propose a fine bands subset selection to obtain the minimum number of strongly discriminative bands. In most cases, scholars focus on different numbers of selected bands to explain the influence on accuracy measures, neglecting the consideration of how many bands to choose is appropriate. Considering this problem, inspired by the traditional image signal-to-noise ratio, we propose MBTBR, which is an automatic strategy to obtain the minimum number of strongly discriminative bands. Through the calculation of MBTBR for the coarse HSI bands subset, we further obtain the fine HSI bands subset.

To obtain the minimum number of recommended bands whose AD results are approximately equal to the results of all spectral bands, MBTBR further measures the redundancy of coarse selected bands. The benchmark algorithm for HAD is RX [12], which assumes that the background can be modeled by a Gaussian multivariate distribution. Inspired by this method, the background in our method is modeled by a Gaussian multivariate distribution, which can be represented by the mean vector and covariance matrix of whole image spectral. The anomaly target is characterized by calculating the Mahalanobis distance between the test pixel and background. MBTBR is a normalized metric that quantifies the separation of the mean value of anomaly target and the mean value of background in N-dimensional space with respect to the variance of the background distribution. MBTBR represents the difference in radiated energy between the anomaly target and the background for multiband images, defined as follows:

$$\mathrm{MBTBR} = \sqrt{(\mu_{\mathrm{target}} - \mu_{\mathrm{back}})^T \sum^{-1} (\mu_{\mathrm{target}} - \mu_{\mathrm{back}})}$$
$$\sum = \frac{1}{M} \sum_{i=1}^{M} (x_i - \mu_{\mathrm{back}})(x_i - \mu_{\mathrm{back}})^T \tag{9}$$

where μ_{target} is the anomaly target spectral mean vector, μ_{back} is the background spectral mean vector, and $\sum$ is the covariance matrix of background pixels. The aim of calculating MBTBR is to determine the recommended number of bands, which is the inflection point. According to (9), we select the band with the highest MBTBR value as the strongly discriminative band. Then, we obtain the strongly discriminative hyperspectral subimage $Z = \{x_1, x_2, \ldots, x_n\} \in \mathbb{R}^{WH \times n}$ containing these n recommended bands with size of $WH \times n$.

C. Adaptive Spatial–Spectral Feature Extraction Strategy

This section presents a novel adaptive spatial–spectral feature extraction strategy that takes into account the characteristics of spatial and spectral anomalies in HSIs. According to the anomaly characteristics in HSIs, spectral anomalies can be regarded as anomalies with different spectral forms from the background. Spatial anomalies consider that anomalous targets are often sparsely distributed in HSIs with a low occurrence probability. This strategy can improve the accuracy of anomaly scoring in HSIs without increasing the computational complexity of the algorithm.

1) Spectral Feature Extraction: In the field of hyperspectral AD, statistic-based methods run fast and satisfy real-time detection. However, these methods excessively rely on the accuracy of background statistics. In hyperspectral imaging, the spectral curve is easily disturbed by various complex factors, such as noise and dark current, causing oscillation in spectral curves. Moreover, in real scenarios, anomalous targets play the role of outliers, contaminating the background data and hindering the accurate estimation of background pixels. Both phenomena could seriously degrade the performance of statistic-based AD methods. Considering this problem, we use

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Process of the Proposed HAD Method

Input: The observed HSI $X \in \mathbb{R}^{WH\times N}$; Number of iterations $iters$.

Output: Final anomaly detection map R_{fin}.

Stage 1: Semi-supervised discriminative band selection strategy.

1. The observed HSI X is coarse partitioned by Eq. (1).
2. Calculate initial cluster center x_{λ_m} of coarse partition groups by Eq. (3).
3. Define a and b are the neighborhood boundaries of initial cluster center and calculated by Eq. (4).
4. Define R to record the similarity between current band which want to be repartitioned x_j and cluster center x_{λ_m}, and initialize it to infinity. Define A to express the regroup label and initialize it to zero.
5. Calculate the Euclidean distance $D(x_{\lambda_m}, x_j)$ between cluster center x_{λ_m} and current band x_j.
6. **for** $i = 1 : iters$ **do**
7. Update R and A by comparing $D(x_{\lambda_m}, x_j)$ and $R(x_j)$.
8. **end**
9. According to A, select the maximum value of the index in each group to update the partition point G.
10. The fine partition strategy of HSI cube is conducted as $X_m = \{x_i\}_{i=G(m)}^{i=G(m+1)}, m = 1, 2, \ldots, M-1$.
11. Calculate spatial local density of each band by Eq. (6).
12. Calculate spectral information entropy of each band by Eq. (7).
13. Obtain the coarse HSI bands subset $Y \in \mathbb{R}^{WH\times M}$ via calculate the weight of each band by Eq. (8).
14. Calculate the MBTBR by Eq. (9) to obtain the strongly discriminative hyperspectral sub-image $Z \in \mathbb{R}^{WH\times n}$.

Stage 2: Adaptive spatial–spectral feature extraction strategy.

15. Extract the spectral feature R_{spe} of Z by Eq. (12).
16. Extract the spatial feature R_{spa} of Z by Eq. (13).
17. Obtain the final anomaly detection map R_{fin} by Eq. (14).

Mahalanobis distance [12] to obtain the initial spectral feature extraction result and utilize saliency weight for further spectral feature enhancement. A larger Mahalanobis distance implies more anomalous pixels. The initial spectral feature extraction result can be described as follows:

$$R_{x-\text{mahal}} = (x_i - \mu_Z)^T \Gamma_Z^{-1} (x_i - \mu_Z), \quad i = 1, 2, \ldots, WH \tag{10}$$

where x_i is the ith pixel in the hyperspectral subimage Z with n bands, and μ_Z and Γ_Z are the mean vector and covariance matrix of Z. We perform saliency enhancement on the above spectral feature. Considering the influence of adjacent pixels on the testing pixel, we compute spectral saliency weight in a neighborhood region (two horizontal, two vertical, and the four diagonal), where the region size is $\omega = 3$. The saliency distance D_{sal} between the testing pixel x and any neighborhood pixel y_i contained in the neighborhood region is calculated by

$$D_{\text{sal}}(x, y_i) = \frac{\arccos\left(\frac{x y_i}{\|x\|_2 \|y_i\|_2}\right)}{1 + c \cdot \sqrt{(i-k)^2 + (j-l)^2}}, \quad i = 1, 2, \ldots, w \tag{11}$$

where $w = \omega \times \omega - 1 = 8$ is the number of adjacent pixels contained in the region. c is a constant that could cause little effect on the result, and we set its value to 1. (i, j) and (k, l) are the position coordinate of the pixels x and y_i. The saliency distance is calculated as the ratio of the spectral distance and the spatial position distance. Finally, the saliency weight of testing pixel is obtained by averaging all the saliency weights in the neighborhood region. When the spectral signal of testing pixel is significantly different from that of its adjacent pixels, the pixel is considered regionally salient. Therefore, the final detection result in spectral domain R_{spe} is reformulated as follows:

$$R_{\text{spe}} = R_{x-\text{mahal}} \frac{\sum_{i=1}^{w} D_{\text{sal}}}{w}. \tag{12}$$

2) Spatial Feature Extraction: If the test pixel is abnormal, its spatial structure will differ from the spatial structure of its neighboring background pixels. To calculate the spatial structural similarity between the test pixel and its neighborhood background pixels, we propose a local spatial similarity measurement method. We perform a local partition between the test pixel and its neighborhood background pixels. The test pixel and its surrounding pixels (two horizontal, two vertical, and the four diagonal) are referred to as the center window, and its size is $\omega = 3$. The neighborhood background window is defined as the region which consists of pixels surrounding the center window. The size of the neighborhood background window is also set to 3 to match the size of the central window. By sliding the neighborhood background window clockwise around the center window, we obtain the local spatial similarity information around the center window. The most similar spatial structure window is searched from the background region. The spatial feature between center window and the neighboring background window is calculated by

$$R_{\text{spa}} = \min(\text{tr}(W_x - W_{y_i})^T \times \text{tr}(W_x - W_{y_i})) \quad i = 1, 2, \ldots, 8 \times \omega \tag{13}$$

where W_x denotes the center window on testing pixel x, and W_{y_i} is the neighborhood background window on adjacent pixel y_i. ω is the size of the spatial structure window, in this article, $\omega = 3$. tr() is the matrix trace. Theoretically, spatial structure windows of different sizes and shapes should be chosen to better adapt to the local features of the image. In our method, the neighborhood region of the testing pixel is chosen as the center window. The square with a size of 3 serves as our spatial structure window. In (13), we obtain the spatial structural similarity between the neighborhood region surrounding the test pixel and its neighboring background window. The greater the similarity, the smaller the value, and the more likely it is to be the background, and vice versa.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

3) Adaptive Spectral–Spatial Feature Fusion Strategy: To highlight anomaly targets, two complementary features extracted above are adaptively fused. To this end, we propose an adaptive feature fusion strategy. A linear score function, as shown in (14), is employed as the fusion strategy

$$R_{\text{fin}} = a_{\text{spe}} \times R_{\text{spe}} + b_{\text{spa}} \times R_{\text{spa}} \tag{14}$$

where R_{fin} is the final AD result. a_{spe} and b_{spa} are the corresponding weight factors. It is difficult to empirically tune them to obtain the best value, and a common choice in the existing literature is the average pooling operation, i.e., $a_{\text{spe}} = b_{\text{spa}} = 0.5$. In this article, an adaptive score mechanism is used to adaptively determine the values of a_{spe} and b_{spa}, calculated by

$$\begin{aligned} a_{\text{spe}} &= \frac{\sqrt{\lambda_{e-\max}\left(R_{\text{spe}}^T R_{\text{spe}}\right)}}{\sqrt{\lambda_{e-\max}\left(R_{\text{spe}}^T R_{\text{spe}}\right)} + \sqrt{\lambda_{a-\max}\left(R_{\text{spa}}^T R_{\text{spa}}\right)}} \\ b_{\text{spa}} &= \frac{\sqrt{\lambda_{a-\max}\left(R_{\text{spa}}^T R_{\text{spa}}\right)}}{\sqrt{\lambda_{e-\max}\left(R_{\text{spe}}^T R_{\text{spe}}\right)} + \sqrt{\lambda_{a-\max}\left(R_{\text{spa}}^T R_{\text{spa}}\right)}} \end{aligned} \tag{15}$$

where $\lambda_{e-\max}()$ is the largest eigenvalue of matrix $R_{\text{spe}}^T R_{\text{spe}}$, and $\lambda_{a-\max}()$ is the largest eigenvalue of matrix $R_{\text{spa}}^T R_{\text{spa}}$. By calculating the maximum eigenvalue of spectral and spatial features, its weight in the final AD result is adaptively adjusted. Such adaptive spatial–spectral feature fusion strategy helps to increase the AD performance. The key steps of our proposed HAD method are summarized in Algorithm 1.

IV. Experiments Results and Discussion

In this section, extensive experiments are conducted to evaluate and verify the effectiveness and generalization of the proposed method.

A. Data Description

There are four real HSIs captured over different scenes by different sensors. In these images, some anomalies appear as points, some exhibit structural information, and most anomalies appear in various scales, which are listed as follows.

1) *ABU-Airport Dataset:* The first dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [29]. After removing SNR and water absorption bands, three scenarios in ABU-Airport dataset are of size 100 × 100 with 205 spectral bands, covering the spectral range from 400 to 2500 nm, and the spatial resolution is 7.1 m/pixel. Another scenario is of size 100 × 100 × 191, with a spatial resolution of 3.4 m/pixel. The three airplanes in different scales are anomalous targets of interest, consisting of 60 pixels and accounting for 0.6% of the image.
2) *ABU-Beach Dataset:* The second dataset was collected by the AVIRIS sensor and Reflective Optics System Imaging Spectrometer (ROSIS-03) [29]. After removing SNR and water absorption bands, the first scenario is of size 150 × 150 × 188 with a spatial resolution of 17.2 m. The second scenario is of size 100 × 100 × 193 with a spatial resolution of 7.5 m. The third one is of size 100 × 100 × 188 with a spatial resolution of 4.4 m. The last one is of size 150 × 150 × 102 with a spatial resolution of 1.3 m in wavelengths ranging from 430 to 860 nm. The main background classes are bridge, water, and bare soil. There are some vehicles on the bridge, selected as anomalies. These targets are composed of 63 pixels, accounting for 0.28% of the image.
3) *ABU-Urban Dataset:* The third dataset was collected by the AVIRIS sensor [29]. After removing SNR and water absorption bands, the first scenario is of size 100 × 100 × 204 with a spatial resolution of 17.2 m. The second scenario is of size 100 × 100 × 207 with a spatial resolution of 17.2 m. The third one is of size 100 × 100 × 191 with a spatial resolution of 3.5 m. The last two is of size 100 × 100 × 205 with a spatial resolution of 7.1 m in wavelengths ranging from 400 to 2500 nm. The ABU-Beach dataset source has a relatively simple background, where a large difference exists between the anomaly targets and the background. In contrast, the background in the data sources of ABU-Airport dataset and ABU-Urban dataset is more complicated.
4) *San Diego Dataset:* The fourth dataset is a widely used HSI collected by the AVIRIS sensor over the San Diego airport area, CA, USA. The San Diego-1 image contains 100 × 100 pixels with 189 spectral bands in wavelengths ranging from 400 to 2500 nm [29]. We consider three airplanes as the objects to be detected in this scene, composed of full-pixel anomalies in the main body of the airplanes and subpixel targets on the edges of the airplanes. The background mainly consists of buildings with different roofs, parking aprons with different materials, an airport runway, and a small quantity of vegetation. These targets are composed of 57 pixels, accounting for 0.57% of the image. The San Diego-2 image also contains 100 × 100 pixels, with 189 spectral bands in wavelengths ranging from 400 to 2500 nm. We consider three airplanes as the objects to be detected in this scene. This dataset represents an airport scene in which the main background types are hangars, parking aprons, and exposed soil. The ground truth of this dataset includes 134 anomalous pixels, accounting for 1.34% of the image.

B. Comparison Methods and Evaluation Metrics

To verify the effectiveness of the proposed method, we employed typical methods, the RX [12] detector and the CRD [36] detector, for comparison purposes. Additionally, we compared our method with three state-of-the-art algorithms, CRDBPSW [40], GAED [53], SSFAD [22], and SFBA [60]. The RX method is a representative statistical model-based method, while the CRD and CRDBPSW methods are collaborative representation model-based methods. The GAED method is a deep learning-based method, and the SSFAD method focuses on spatial–spectral feature extraction. The SFBA method utilizes band selection to reduce

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

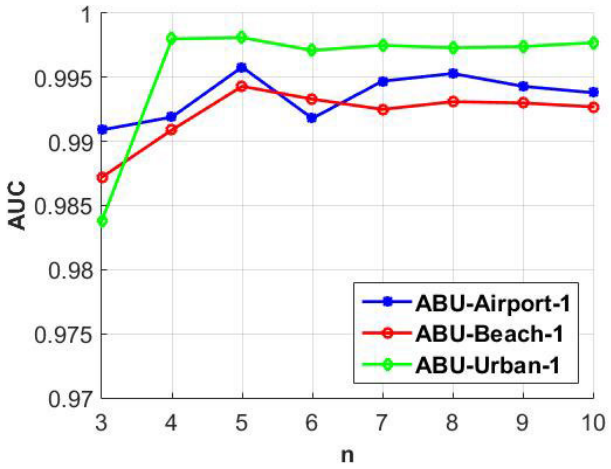


Fig. 4. Impact of the selected band number.

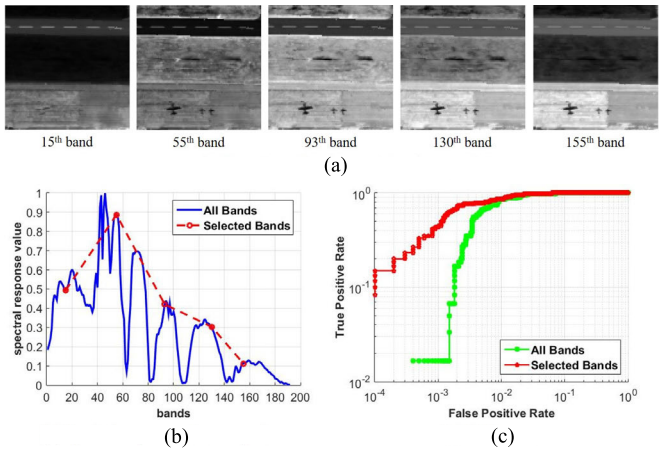


Fig. 5. Selected bands visualization. (a) Selected bands (single-band grayscale image). (b) Selected response value comparison. (c) ROC curves.

the computational complexity. For all compared algorithms, we set the optimal regularization parameters according to the original literature. The execution time of all the experiments is calculated starting with the input as the original HSI to ensure the fairness of experimental running efficiency.

The performance of the proposed method is evaluated using the widely adopted receiver operating characteristic (ROC) curve and the area under the curve (AUC) metric [29], [61], [62], [63]. Originating from communication theory, the ROC curve describes detection performance by plotting the true positive rate (TPR) against the false positive rate (FPR) at each threshold. TPR signifies the number of correct positive detections among all positive samples and FPR denotes the number of incorrect positive detections among all available negative samples. This curve illustrates the tradeoff between TPR and FPR. The ROC curve is a popular criterion for assessing performance in target detection applications. In practical AD applications, the probability of detection (PD) at low probability of false alarm (PFA) is considered the most important criterion, so all ROC curves in this article are plotted in logarithmic coordinate system.

The AUC metric represents the integration of the ROC curve in linear coordinate system, which depends solely on the pixels rather than the absolute detection values [29], [61], [62], [63]. The ideal value of AUC is 1.

C. Parameter Sensitivity Analysis

This section first conducts parameter sensitivity analysis of our proposed method, examining the impact of choosing different band numbers on the accuracy of the HAD problem. The ABU dataset is used for experimentation, and we vary the selected band number from 3 to 10. The resulting AUC values (linear coordinate system) are depicted in Fig. 4, with $n = 3$ serving as the baseline. With the increase in the selected number of bands, the detection accuracy shows a significant improvement. This suggests that as the selected number of bands increases, the spectral information is more abundant, leading to improved AD accuracy. The detection performance reaches the optimum when $n = 5$. Beyond this point, as the selected number of bands exceeds 5, the detection accuracy stabilizes. This indicates that the spectral information provided by five bands is sufficient for robust hyperspectral AD. As the selected number of bands increases, computational complexity rises. Hence, for our experiments, we set the default number of bands to 5.

We also compare the selected bands for the ABU-Airport-1 dataset with full range bands. The selected bands are [15, 55, 93, 130, 155]. The single-band grayscale images are shown in Fig. 5(a). The spectral response value comparison at pixel (35, 85) (anomaly target) is shown in Fig. 5(b). Fig. 5(c) displays the ROC performances (logarithmic coordinate system). The proposed band selection strategy in this article can accurately extract strong discriminative bands, effectively reduce the computational complexity, and guarantee excellent detection accuracy.

D. Ablation Analysis

We achieve fast detection by employing discriminative band selection without compromising the accuracy of AD. The discriminative band selection model utilizes band selection to achieve rapid detection, while the adaptive spatial–spectral feature extraction model ensures detection accuracy. To verify their effectiveness individually, we conduct experiments under six scenarios: 1) without band selection and only with spectral feature (denoted by "No Band Sel-Spe"); 2) without band selection and only with spatial feature (denoted by "No Band Sel-Spa"); 3) without band selection and with spectral–spatial feature (denoted by "No Band Sel-SpeSpa"); 4) band selection and only with spectral feature (denoted by "'Band Sel-Spe"); 5) band selection and only with spatial feature (denoted by 'Band Sel-Spa"); and 6) band selection and with spectral–spatial feature (denoted by "Band Sel-SpeSpa"). We use the ABU-Airport-1 dataset for this ablation study while keeping other settings the same.

Table I presents the AUC metrics and running time under these six scenarios. The execution time is timed from the beginning when the input is the original HSI. Fig. 6 visually illustrates the detection results. Fig. 7(a) displays the ROC performances. The results in Table I indicate that the proposed method obtains poor results without spatial feature extraction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

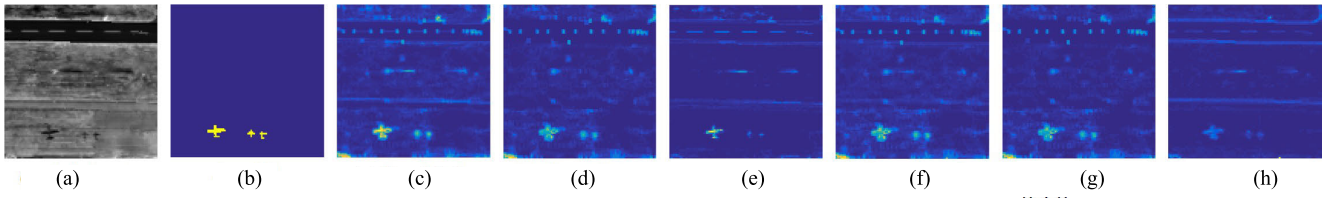


Fig. 6. Detection maps for ablation study. (a) 45th band. (b) Ground-truth map. (c) Band Sel-SpeSpa. (d) Band Sel-Spa. (e) Band Sel-Spe. (f) No Band Sel-SpeSpa. (g) No Band Sel-Spa. (h) No Band Sel-Spe.

TABLE I

AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIMES FOR ABLATION STUDY. THE TOP TWO VALUES ARE HIGHLIGHTED BY BOLD RED AND BLUE

Method	AUC ↑	Time(s) ↓
No Band Sel-Spe	0.9921	6.0204
No Band Sel-Spa	0.9756	6.2178
No Band Sel-SpeSpa	**0.9929**	6.2114
Band Sel-Spe	0.9924	3.2645
Band Sel-Spa	0.9702	**3.2314**
Band Sel-SpeSpa	**0.9958**	**3.0875**

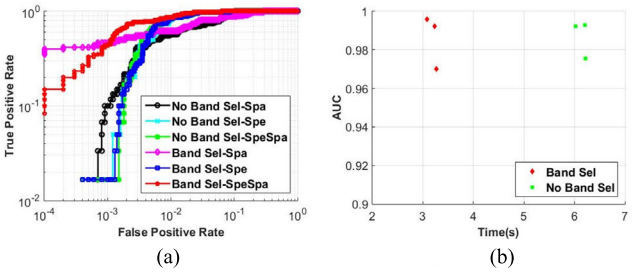


Fig. 7. Analysis of ablation experiments. (a) ROC curves (logarithmic coordinate system). (b) Execution times.

but achieves superior results with spatial–spectral features, demonstrating the effectiveness of spatial features. Omitting spatial features in the proposed method leads to a significant drop in AUC value (Table I). It is known that anomalies in HSIs should exhibit two properties: a spectral difference from the background objects and a lower abundance compared to the majority of the background object pixels. Spatial features play a crucial role in enhancing the distinguishability between the object and the background. The results further validate the importance of spatial features in improving detection outcomes. Table I highlights that after discriminative band selection, the running speed is greatly improved without compromising detection accuracy. Fig. 7(b) illustrates that the execution time of this algorithm is reduced by half after band selection. Consequently, the proposed method achieves fast and accurate anomaly target detection.

In order to verify the effectiveness of the adaptive fusion strategy of spatial and spectral features, we conduct experiments under adaptive fusion and average fusion ($a_{\text{spe}} = b_{\text{spa}} = 0.5$). Fig. 8 visually illustrates the detection results. Fig. 8(e) displays the ROC performances. Table II presents the AUC metrics and running time. Adaptively adjusting the weights of spatial and spectral features in the final AD results can effectively improve the AD performance.

TABLE II

AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIMES

Method	AUC ↑	Time(s) ↓
Average	0.9924	3.8471
Adaptive	**0.9958**	**3.0875**

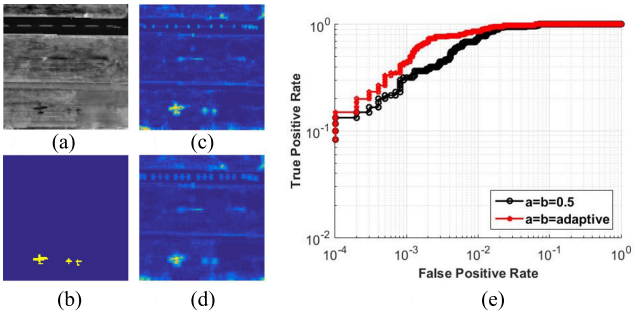


Fig. 8. Detection maps and ROC curves (logarithmic coordinate system). (a) 45th band. (b) Ground-truth map. (c) Adaptive. (d) Average. (e) ROC curves.

E. Detection Results on ABU Datasets

For the ABU datasets, reference maps and visual detection results are displayed in Figs. 9, 11, and 13. The proposed method exhibits the smallest visual difference between the reference and detection maps when compared with the RX, CRD, and GAED methods. In the case of ABU-Airport and ABU-Urban datasets, CRDBPSW and CRD methods struggle to identify abnormal targets. Our proposed algorithm, in comparison with the SSFAD method, demonstrates a superior background suppression effect. The background suppression of the SFBA method is poor. It is impossible to distinguish the abnormal target from the background. Specifically, our proposed method successfully detects anomalies in backgrounds of varying complexity, as observed in the second and fourth rows of Figs. 9 and 11, as well as the third and fifth rows in Fig. 13. In scenarios with complex backgrounds, compared methods fail to accurately reconstruct the background, rendering them ineffective. Across backgrounds with varying complexities, such as airport, beach, and urban scenes, our proposed algorithm maintains stable AD accuracy.

With the best parameters, Figs. 10, 12, and 14 show the ROC performance of different anomaly detectors for ABU datasets. The AUC values are demonstrated in Tables III–V. From the ROC curves, the proposed method outperforms RX, CRD, CRDBPSW, GAED, SSFAD, and SFBA, as the curves of the detection results consistently remain above those of the compared methods. Our proposed method has detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

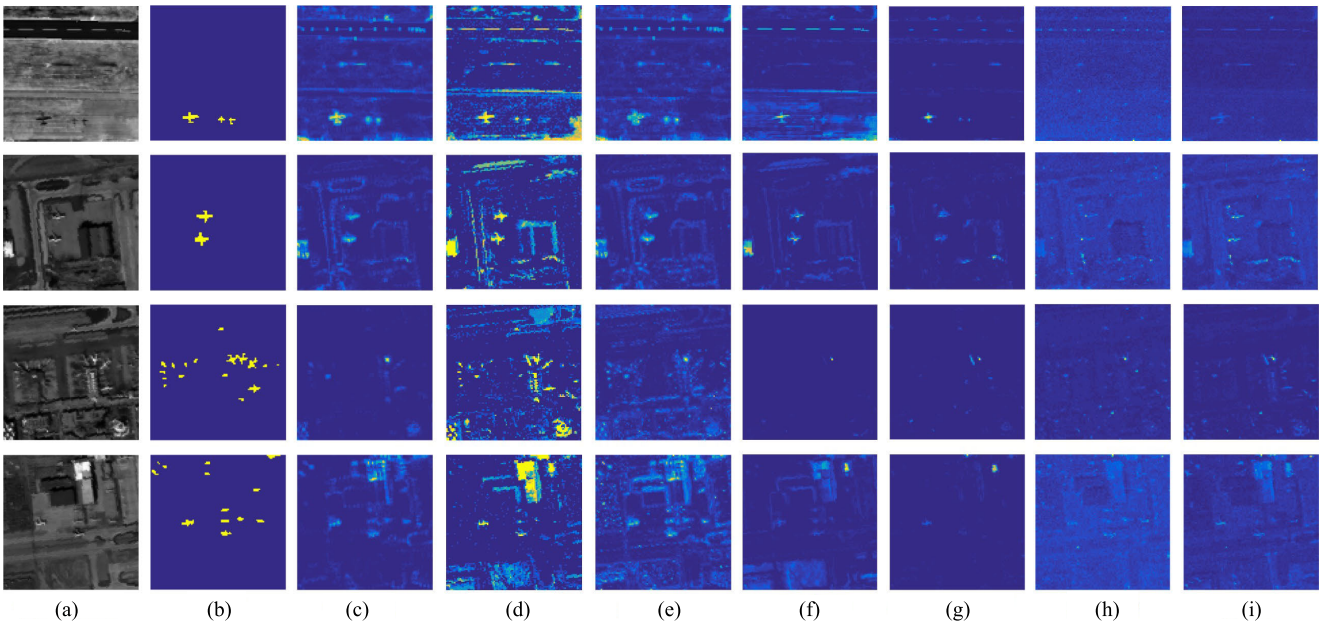


Fig. 9. Detection maps of various anomaly detectors for ABU-Airport dataset. (a) 45th band. (b) Ground-truth map. (c) Ours. (d) SFBA. (e) SSFAD. (f) GAED. (g) CRDBPSW. (h) CRD. (i) RX.

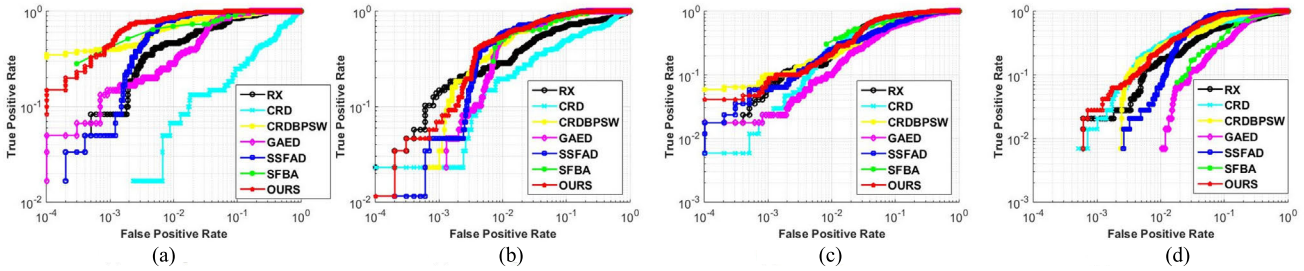


Fig. 10. ROC curves for ABU-Airport dataset (logarithmic coordinate system). (a) ABU-Airport-1. (b) ABU-Airport-2. (c) ABU-Airport-3. (d) ABU-Airport-4.

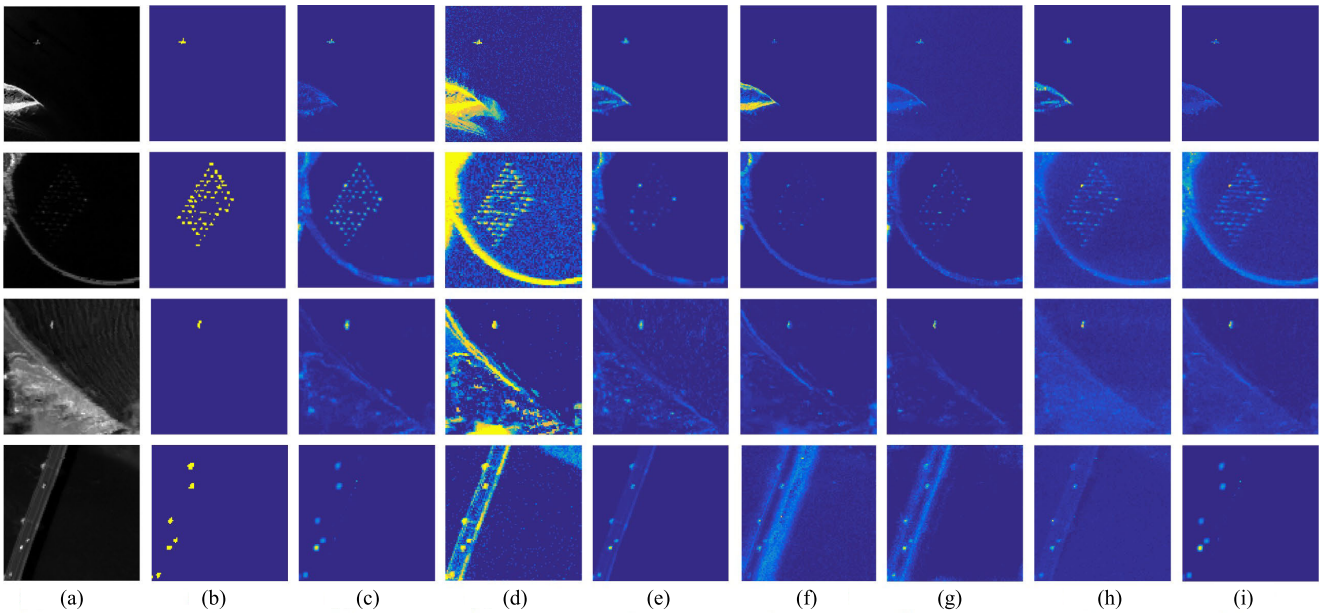


Fig. 11. Detection maps of various anomaly detectors for ABU-Beach dataset. (a) 45th band. (b) Ground-truth map. (c) Ours. (d) SFBA. (e) SSFAD. (f) GAED. (g) CRDBPSW. (h) CRD. (i) RX.

probability at low FPR (in the 1e–4 to 1e–2 region), which can guarantee the detection accuracy in practice. The SFBA method has the worst ROC curve, which fails to detect at low FPR (less than 1e–2). The AUC of the proposed method also achieves excellent detection effectiveness on these experimental datasets. Even for ABU-Airport and ABU-Urban datasets,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

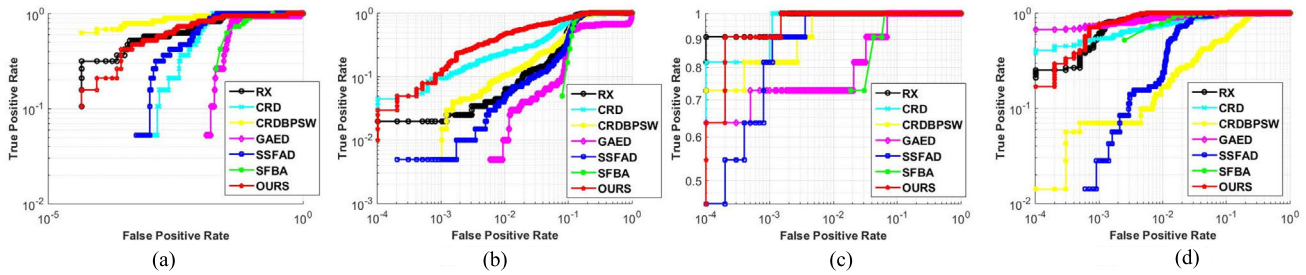


Fig. 12. ROC curves for ABU-Beach dataset (logarithmic coordinate system). (a) ABU-Beach-1. (b) ABU-Beach-2. (c) ABU-Beach-3. (d) ABU-Beach-4.

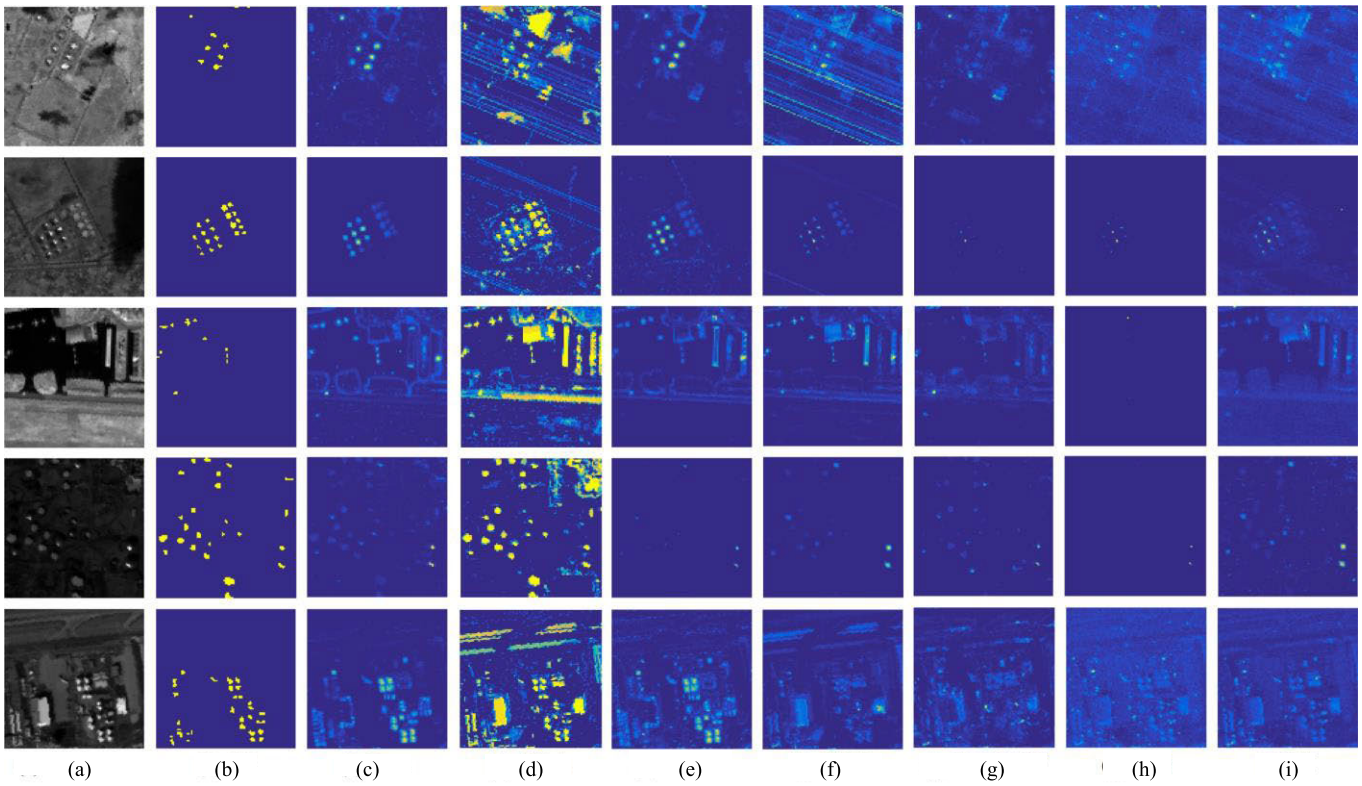


Fig. 13. Detection maps of various anomaly detectors for ABU-Urban dataset. (a) 45th band. (b) Ground-truth map. (c) Ours. (d) SFBA. (e) SSFAD. (f) GAED. (g) CRDBPSW. (h) CRD. (i) RX.

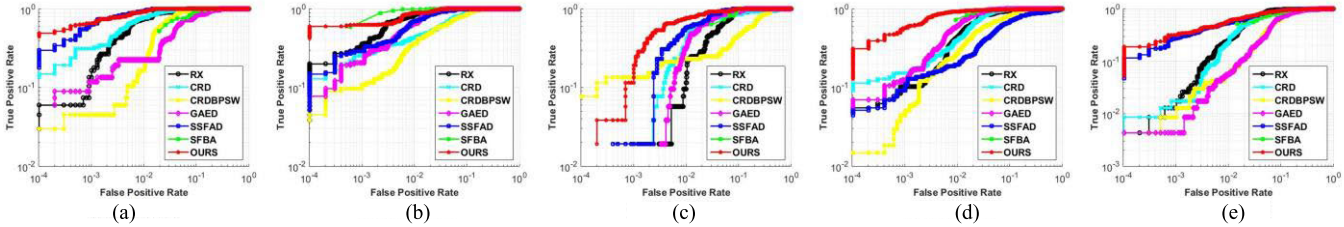


Fig. 14. ROC curves for ABU-Urban dataset (logarithmic coordinate system). (a) ABU-Urban-1. (b) ABU-Urban-2. (c) ABU-Urban-3. (d) ABU-Urban-4. (e) ABU-Urban-5.

having complex backgrounds, the ROC curves and AUC values of our proposed method indicate the best-performing detection result. In general, the ROC curves of the detection results and the AUC values reported in Tables III–V confirm that our method yields promising results across different scenes. This affirms the algorithm's robustness and applicability to a wide range of scenarios.

Tables III–V show the speed of HAD algorithms under the same computing environment. Our proposed algorithm exhibits faster execution speed while ensuring high detection accuracy. None of these algorithms achieve real-time detection due to the high computational complexity of the existing high-accuracy detection algorithms. Despite RX and SFBA exhibiting a shorter running time, its low detection accuracy must not be neglected. The computational complexity of RX can be expressed as $O(N^3) + O(N^2)$. However, when our proposed HAD algorithm, based on critical band selection, is considered, the computational complexity is only

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

TABLE III

AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIME OF VARIOUS ANOMALY DETECTORS FOR ABU-AIRPORT DATASET. THE TOP THREE VALUES ARE HIGHLIGHTED BY BOLD RED, BLUE, AND BLACK

Metric	Dataset	RX 1990 [12]	CRD 2014 [36]	CRDBPSW 2022 [40]	GAED 2023 [53]	SSFAD 2023 [22]	SFBA 2022 [60]	OURS
AUC ↑	Airport-1	0.9526	0.6274	0.9743	0.9678	**0.9930**	**0.9758**	**0.9958**
	Airport-2	0.8404	0.7658	0.9637	**0.9690**	**0.9728**	0.9394	**0.9750**
	Airport-3	**0.9288**	**0.9312**	0.8971	0.8630	0.8954	0.8968	**0.9478**
	Airport-4	0.8221	0.8835	**0.9237**	0.8154	**0.9518**	0.7546	**0.9613**
Time(s) ↓	Airport-1	**0.4897**	7.3329	298.8121	73.1218	60.6456	**1.6867**	**3.0875**
	Airport-2	**0.5331**	8.8062	372.5271	74.3662	55.5217	**1.5891**	**3.6541**
	Airport-3	**0.5592**	7.6972	345.8669	75.9554	59.4453	**1.5899**	**3.7614**
	Airport-4	**0.5252**	8.4024	372.4341	76.4571	58.9046	**1.5746**	**3.6814**

TABLE IV

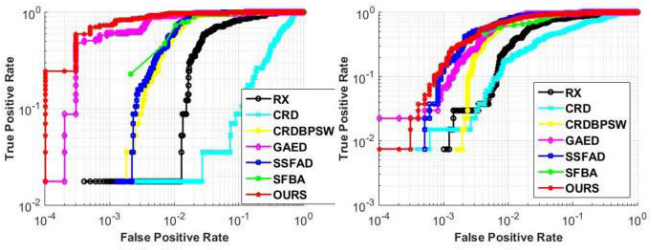
AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIME OF VARIOUS ANOMALY DETECTORS FOR ABU-BEACH DATASET. THE TOP THREE VALUES ARE HIGHLIGHTED BY BOLD RED, BLUE, AND BLACK

Metric	Dataset	RX 1990 [12]	CRD 2014 [36]	CRDBPSW 2022 [40]	GAED 2023 [53]	SSFAD 2023 [22]	SFBA 2022 [60]	OURS
AUC ↑	Beach-1	0.9807	0.9703	**0.9949**	0.9223	**0.9933**	0.9579	**0.9943**
	Beach-2	**0.9106**	**0.9269**	0.9020	0.6163	0.8969	0.8820	**0.9603**
	Beach-3	**0.9998**	**0.9998**	0.9993	0.9888	0.9993	0.9810	**0.9998**
	Beach-4	**0.9934**	0.9887	0.9034	**0.9948**	0.9918	0.9928	**0.9990**
Time(s) ↓	Beach-1	**2.4929**	15.8925	962.0304	140.8092	96.1842	**1.4081**	**6.3087**
	Beach-2	**0.4973**	8.1887	364.7844	71.0897	47.6405	**1.5812**	**3.6369**
	Beach-3	**0.4993**	7.9861	387.0232	61.9421	46.4231	**1.5586**	**3.6296**
	Beach-4	**0.3641**	5.3875	281.2731	43.8302	21.2435	**1.5572**	**2.7247**

TABLE V

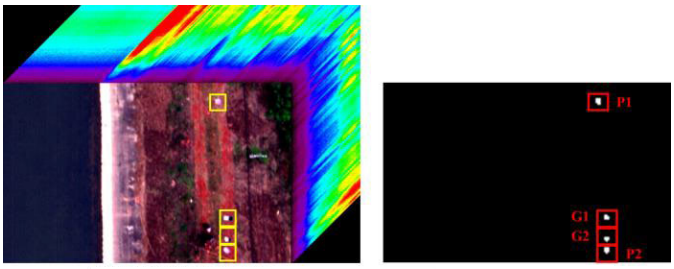
AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIME OF VARIOUS ANOMALY DETECTORS FOR ABU-URBAN DATASET. THE TOP THREE VALUES ARE HIGHLIGHTED BY BOLD RED, BLUE, AND BLACK

Metric	Dataset	RX 1990 [12]	CRD 2014 [36]	CRDBPSW 2022 [40]	GAED 2023 [53]	SSFAD 2023 [22]	SFBA 2022 [60]	OURS
AUC ↑	Urban-1	**0.9907**	0.9917	0.9755	0.9431	**0.9980**	0.9642	**0.9981**
	Urban-2	**0.9946**	0.9645	0.9560	0.9917	0.9890	**0.9989**	**0.9958**
	Urban-3	0.9513	0.9406	0.8491	0.9728	**0.9837**	**0.9647**	**0.9897**
	Urban-4	0.9730	0.9779	0.9508	**0.9887**	0.8661	**0.9879**	**0.9925**
	Urban-5	**0.9692**	0.9569	0.9028	0.9066	**0.9732**	0.9471	**0.9737**
Time(s) ↓	Urban-1	**0.5359**	8.4637	379.4736	63.0023	47.4768	**1.5591**	**3.1297**
	Urban-2	**0.5084**	8.2922	380.6669	68.3116	54.4371	**1.7495**	**3.5928**
	Urban-3	**0.4879**	8.1601	380.2052	70.8037	48.5003	**1.5901**	**3.6441**
	Urban-4	**0.5218**	8.3835	423.4544	64.4245	53.4970	**0.9634**	**3.5722**
	Urban-5	**0.5075**	8.4097	333.1486	65.3362	53.9193	**1.5834**	**3.6247**



(a) (b)

Fig. 15. ROC curves for San Diego dataset (logarithmic coordinate system). (a) San Diego-1. (b) San Diego-2.



(a) (b)

Fig. 16. WHU-Hi-River dataset. (a) RGB pseudo-color image. (b) Ground-truth map.

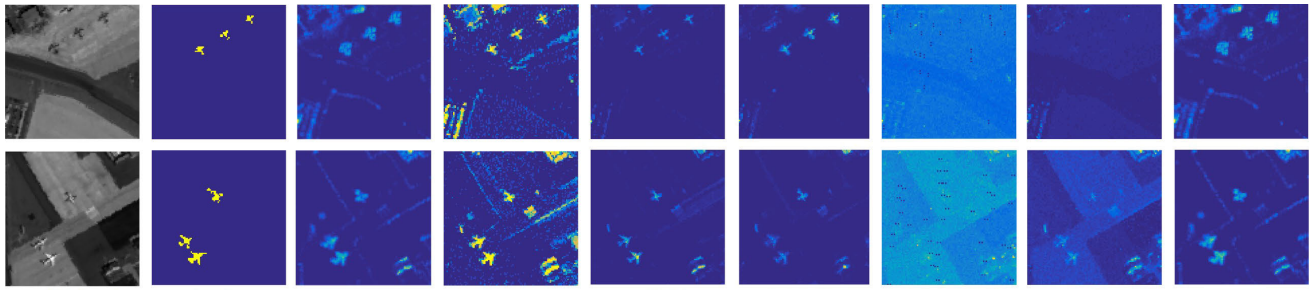
$O(n^3)+O(n^2)+O(2nM)$. In this expression, N is the number of spectral bands in HSI, n is the number of selected bands, and $M = HW$ is the number of pixels in HSIs. Usually, the number of bands N in the HSIs can reach several hundreds, whereas n of the proposed algorithm is greatly reduced compared to N. Therefore, the proposed method can achieve

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIME OF VARIOUS ANOMALY DETECTORS FOR SAN DIEGO DATASET. THE TOP THREE VALUES ARE HIGHLIGHTED BY BOLD RED, BLUE, AND BLACK

Metric	Dataset	RX **1990** [12]	CRD **2014** [36]	CRDBPSW **2022** [40]	GAED **2023** [53]	SSFAD **2023** [22]	SFBA **2022** [60]	OURS
AUC ↑	San Diego-1	0.9398	0.6151	0.9893	**0.9956**	**0.9905**	0.9794	**0.9977**
	San Diego-1	0.9401	0.8411	**0.9863**	**0.9883**	0.9858	0.9765	**0.9886**
Time(s) ↓	San Diego-1	**0.4845**	7.7268	299.2531	60.2009	51.9003	**1.5838**	**3.1682**
	San Diego-1	**0.4701**	7.0807	303.0531	62.0484	45.9181	**1.5689**	**3.2794**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 17. Detection maps of various anomaly detectors for San Diego dataset. (a) 45th band. (b) Ground-truth map. (c) Ours. (d) SFBA. (e) SSFAD. (f) GAED. (g) CRDBPSW. (h) CRD. (i) RX.

TABLE VII

AUC VALUES (LINEAR COORDINATE SYSTEM) AND EXECUTION TIME OF VARIOUS ANOMALY DETECTORS FOR WHU-HI-RIVER DATASET. THE TOP THREE VALUES ARE HIGHLIGHTED BY BOLD RED, BLUE AND BLACK

Metric	Dataset	RX **1990** [12]	CRD **2014** [36]	CRDBPSW **2022** [40]	GAED **2023** [53]	SSFAD **2023** [22]	SFBA **2022** [60]	OURS
AUC ↑	WHU-Hi-River	**0.9968**	0.9890	0.9966	0.9719	**0.9981**	0.9784	**0.9996**
Time(s) ↓	WHU-Hi-River	**1.2337**	10.6899	458.2233	88.6736	44.2393	**1.8196**	**5.5823**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 18. Detection maps of various anomaly detectors for WHU-Hi-River dataset. (a) 45th band. (b) Ground-truth map. (c) Ours. (d) SFBA. (e) SSFAD. (f) GAED. (g) CRDBPSW. (h) CRD. (i) RX.

real-time anomaly target detection, representing another major advantage.

F. Detection Results on San Diego Dataset

For the San Diego dataset, the reference maps and visual detection results are shown in Fig. 17. The proposed method demonstrates the smallest visual difference between the reference and detection maps compared to RX, CRD, and CRDBPSW methods. In the case of the San Diego dataset, the CRDBPSW method cannot identify abnormal targets. The SFBA method is unable to distinguish between the target and the background. This method becomes ineffective in accurately reconstructing the background in complex scenarios. In comparison to other existing algorithms, our proposed algorithm maintains stable AD accuracy. Fig. 15 illustrates the ROC performance of different anomaly detectors for the San Diego datasets, and the AUC values are presented in Table VI. The SFBA method has the worst ROC curve, which fails to detect at low FPR (less than 1e–2). Table VI also details the speed of HAD algorithm. In comparison with the existing algorithms, our proposed method demonstrates excellent detection result and faster running times, verifying its strong robustness and low computational complexity simultaneously.

G. Additional Public Dataset

To further assess the performance of the proposed method, experiments on another new public dataset have been conducted. The public dataset used is WHU-Hi-River dataset [52], presented in Fig. 16. This data collection was carried out on March 21, 2018, capturing a long river bank scene in Honghu, Hubei province of China. The spatial resolution of the imagery is 6 cm/pixel, and the image scene is of size 105 × 168 pixels with 135 spectral bands in wavelengths ranging from 400 to 1000 nm, as depicted in Fig. 16(a). In the AD task, two plastic plates and two gray panels are considered as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

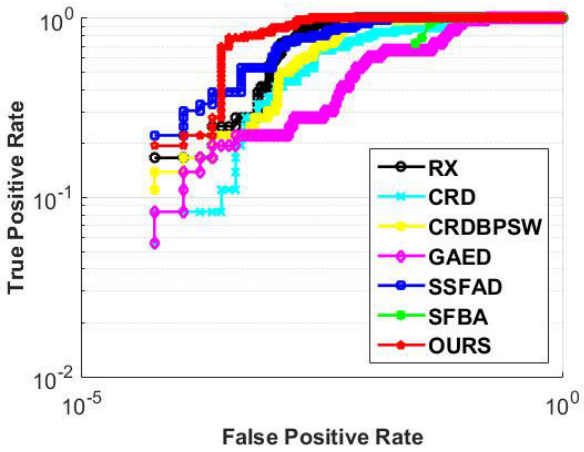


Fig. 19. ROC curves for WHU-Hi-River dataset (logarithmic coordinate system).

anomalies, with a total of 36 pixels, accounting for 0.2% of the image.

For the WHU-Hi-River dataset, the reference map and detection results are visually presented in Fig. 18. The proposed method exhibits the smallest visual difference between the reference and detection maps compared with RX, CRD, CRDBPSW, GAED, and SFBA methods. Notably, our proposed method also presents the most effective background suppression. Fig. 19 illustrates the ROC performance of different anomaly detectors for the WHU-Hi-River dataset, the AUC values detailed in Table VII. Table VII also shows the computational speed of the HAD algorithms under the same computing environment. Compared to the existing algorithms, our proposed method showcases excellent results on the WHU-Hi-River dataset along with faster running times, confirming its robustness and low computational complexity.

V. Conclusion

This article proposes a rapid and robust hyperspectral AD algorithm through discriminative band selection for HSIs. We introduce an adaptively spatial–spectral feature extraction strategy to ensure detection accuracy. To effectively mine contextual information across a broad spectrum, we segment the HSI cube into several groups using a coarse-to-fine partition strategy. Based on spatial local density and spectral information entropy, we identify the most relevant and informative bands to obtain the coarse HSI bands subset. To further enhance detection speed, an MBTBR is designed to select the strongly discriminative band and obtain the fine HSI bands subset. Our proposed adaptive spatial–spectral feature extraction strategy combines spectral features enhanced by saliency weight and spatial features enhanced by fully extracted similarity information from local patches. Experimental results on five real hyperspectral datasets confirm that the proposed method exhibits superior background suppression ability, excellent detection accuracy, and satisfactory speed compared to the state-of-the-art algorithms, validating its strong robustness and low computational complexity simultaneously.

References

[1] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609413.

[2] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517916, doi: 10.1109/TGRS.2021.3128764.

[3] F. Luo, H. Huang, Z. Ma, and J. Liu, "Semisupervised sparse manifold discriminative analysis for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6197–6211, Oct. 2016.

[4] J. Hu, Y. Zhang, M. Zhao, and P. Li, "Spatial–spectral extraction for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[5] X. Shang, M. Song, Y. Wang, and H. Yu, "Residual-driven band selection for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[6] B. Du and L. Zhang, "A discriminative metric learning based anomaly detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6844–6857, Nov. 2014.

[7] A. Taghipour and H. Ghassemian, "Hyperspectral anomaly detection using attribute profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1136–1140, Jul. 2017.

[8] B. Sun, X. Kang, S. Li, and J. A. Benediktsson, "Random-Walker-based collaborative learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 212–222, Jan. 2017.

[9] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[10] X L. Ni and T. Jiang, "Approach study of remote sensing image classification by automatically extracting the optimal bands," *Sci. Surveying Mapping*, vol. 33, no. S1, pp. 89–91, 2008.

[11] Q. Wang, Q. Li, and X. Li, "Hyperspectral band selection via adaptive subspace partition strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4940–4950, Dec. 2019.

[12] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Sep. 1990.

[13] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022.

[14] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[15] S. Matteoli, M. Diani, and G. Corsini, "Hyperspectral anomaly detection with kurtosis-driven local covariance matrix corruption mitigation," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 532–536, May 2011.

[16] J. E. Fowler and Q. Du, "Anomaly detection and reconstruction from random projections," *IEEE Trans. Image Process.*, vol. 21, no. 1, pp. 184–195, Jan. 2012.

[17] Y. Gu, Y. Liu, and Y. Zhang, "A selective KPCA algorithm based on high-order statistics for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 1, pp. 43–47, Jan. 2008.

[18] S. Chang, B. Du, and L. Zhang, "A subspace selection-based discriminative forest method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4033–4046, Jun. 2020.

[19] Y. Ma, G. Fan, Q. Jin, J. Huang, X. Mei, and J. Ma, "Hyperspectral anomaly detection via integration of feature extraction and background purification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 8, pp. 1436–1440, Aug. 2021.

[20] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[21] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.

[22] Z. Hou, S. Cheng, and T. Hu, "A spectral–spatial fusion anomaly detection method for hyperspectral imagery," 2022, *arXiv:2202.11889*.

[23] S. B. Serpico and L. Bruzzone, "A new search algorithm for feature selection in hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1360–1367, Jul. 2001.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.

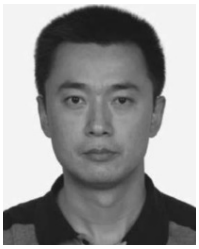
[24] I. Guyon and A. Elisseeff, "An introduction to variable and feature selection," *J. Mach. Learn. Res.*, vol. 3, pp. 1157–1182, Jan. 2003.
[25] M. Li, W. Li, Y. Liu, Y. Huang, and G. Yang, "Adaptive mask sampling and manifold to Euclidean subspace learning with distance covariance representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508518.
[26] L. Wang, H.-C. Li, B. Xue, and C.-I. Chang, "Constrained band subset selection for hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2032–2036, Nov. 2017.
[27] H.-F. Yan, Y.-Q. Zhao, J. C.-W. Chan, and S. G. Kong, "Spectral super-resolution based on dictionary optimization learning via spectral library," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400216, doi: 10.1109/TGRS.2022.3229439.
[28] T. Lu, S. Li, L. Fang, Y. Ma, and J. A. Benediktsson, "Spectral–spatial adaptive sparse representation for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 373–385, Jan. 2016.
[29] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.
[30] J. Yin, Y. Wang, and Z. Zhao, "Optimal band selection for hyperspectral image classification based on inter-class separability," in *Proc. Symp. Photon. Optoelectronics*, Jun. 2010, pp. 1–4.
[31] S. Li, Z. Liu, L. Fang, and Q. Li, "Block diagonal representation learning for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509213, doi: 10.1109/TGRS.2023.3266811.
[32] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.
[33] J.-F. Cardoso, "Dependence, correlation and Gaussianity in independent component analysis," *J. Mach. Learn. Res.*, vol. 4, pp. 1177–1203, Jan. 2003.
[34] Q. Wang et al., "Unsupervised hyperspectral band selection via structure-conserved and neighborhood-grouped evolutionary algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522319, doi: 10.1109/TGRS.2023.3309830.
[35] Q. Wang, Q. Li, and X. Li, "A fast neighborhood grouping method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5028–5039, Jun. 2021.
[36] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.
[37] M. Vafadar and H. Ghassemian, "Hyperspectral anomaly detection using outlier removal from collaborative representation," in *Proc. 3rd Int. Conf. Pattern Recognit. Image Anal. (IPRIA)*, Apr. 2017, pp. 13–19.
[38] H. Su, Z. Wu, Q. Du, and P. Du, "Hyperspectral anomaly detection using collaborative representation with outlier removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 5029–5038, Dec. 2018.
[39] K. Tan, Z. Hou, F. Wu, Q. Du, and Y. Chen, "Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation," *Remote Sens.*, vol. 11, no. 11, p. 1318, Jun. 2019.
[40] Z. Hou, W. Li, R. Tao, P. Ma, and W. Shi, "Collaborative representation with background purification and saliency weight for hyperspectral anomaly detection," *Sci. China Inf. Sci.*, vol. 65, no. 1, Jan. 2022, Art. no. 112305.
[41] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1–37, 2011.
[42] G. Liu, Z. Lin, S. Yan, J. Sun, Y. Yu, and Y. Ma, "Robust recovery of subspace structures by low-rank representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 171–184, Jan. 2013.
[43] Y. Xu, Z. Wu, Z. Wei, H. Liu, and X. Xu, "A novel hyperspectral image anomaly detection method based on low rank representation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4444–4447.
[44] Y. Niu and B. Wang, "Hyperspectral anomaly detection based on low-rank representation and learned dictionary," *Remote Sens.*, vol. 8, no. 4, p. 289, Mar. 2016.
[45] K. Tan, Z. Hou, D. Ma, Y. Chen, and Q. Du, "Anomaly detection in hyperspectral imagery based on low-rank representation incorporating a spatial constraint," *Remote Sens.*, vol. 11, no. 13, p. 1578, Jul. 2019.
[46] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Simultaneous joint sparsity model for target detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 676–680, Jul. 2011.
[47] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.
[48] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.
[49] W. Xie, J. Lei, B. Liu, Y. Li, and X. Jia, "Spectral constraint adversarial autoencoders approach to feature representation in hyperspectral anomaly detection," *Neural Netw.*, vol. 119, pp. 222–234, Nov. 2019.
[50] C. Zhao and L. Zhang, "Spectral–spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 92, pp. 166–176, Aug. 2018.
[51] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Mar. 2020.
[52] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.
[53] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818, doi: 10.1109/TGRS.2022.3207165.
[54] R. Liu, H. Wang, and X. Yu, "Shared-nearest-neighbor-based clustering by fast search and find of density peaks," *Inf. Sci.*, vol. 450, pp. 200–226, Jun. 2018.
[55] W. Rao, Y. Qu, L. Gao, X. Sun, Y. Wu, and B. Zhang, "Transferable network with Siamese architecture for anomaly detection in hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102669.
[56] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.
[57] B. Tu, Z. Wang, H. Ouyang, X. Yang, J. Li, and A. Plaza, "Hyperspectral anomaly detection using the spectral–spatial graph," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542814, doi: 10.1109/TGRS.2022.3217329.
[58] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.
[59] B. Tu, X. Yang, W. He, J. Li, and A. Plaza, "Hyperspectral anomaly detection using reconstruction fusion of quaternion frequency domain analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 8358–8372, Jun. 2024, doi: 10.1109/TNNLS.2022.3227167.
[60] S. Yuan, L. Shi, B. Yao, F. Li, and Y. Du, "A hyperspectral anomaly detection algorithm using sub-features grouping and binary accumulation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[61] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.
[62] X. Hu et al., "Hyperspectral anomaly detection using deep learning: A review," *Remote Sens.*, vol. 14, no. 9, p. 1973, Apr. 2022.
[63] B. Qu, X. T. Zheng, X. M. Qian, and X. Q. Lu, "Research progress on hyperspectral anomaly detection," *Nat. Remote Sens. Bull.*, vol. 28, no. 1, pp. 42–54, 2024.
[64] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. J. Plaza, "PDB-SNet: Pixel-shuffle downsampling blind-spot reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511914.
[65] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531916, doi: 10.1109/TGRS.2023.3335484.
[66] D. Wang, L. Zhuang, L. Gao, X. Sun, X. Zhao, and A. Plaza, "Sliding dual-window-inspired reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504115, doi: 10.1109/TGRS.2024.3351179.



Hao-Fang Yan received the M.S. degree in control science and engineering from Guilin University of Electronic Technology, Guilin, China, in 2019. She is currently pursuing the Ph.D. degree with the School of Automation, Northwestern Polytechnic University, Xi'an, China.

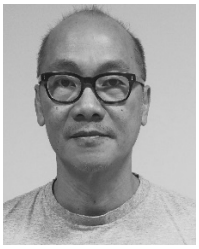
Her research interests include hyperspectral image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.



Yong-Qiang Zhao (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in control science and engineering from Northwestern Polytechnic University, Xi'an, China, in 1998, 2001, and 2004, respectively.

From 2007 to 2009, he was a Post-Doctoral Researcher at McMaster University, Hamilton, ON, Canada, and Temple University, Philadelphia, PA, USA. He is currently a Professor with Northwestern Polytechnical University and the Research and Development Institute, Northwestern Polytechnical University, Shenzhen, China. His research interests include polarization vision, hyperspectral imaging, image processing, and pattern recognition.



Jonathan Cheung-Wai Chan received the Ph.D. degree from The University of Hong Kong, Hong Kong, in 1999.

From 1998 to 2001, he was a Research Scientist with the Department of Geography, University of Maryland, College Park, MD, USA. From 2001 to 2005, he was with the Interuniversity Micro-Electronics Center (imec), Leuven, Belgium. From 2005 to 2011, he worked at the Geography Department, Vrije Universiteit Brussel (VUB), Brussels, Belgium. From 2013 to 2014, he was a Marie Curie Fellow with the Fondazione Edmund Mach, San Michele all'Adige, Italy. He is currently a Senior Researcher and a Guest Professor with the Department of Electronics and Informatics (ETRO), VUB. His research interests include land-cover classification with machine learning algorithms, detailed mapping using hyperspectral data, and spatial and spectral enhancement of satellite hyperspectral images.



Seong G. Kong (Senior Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from Seoul National University, Seoul, South Korea, in 1982 and 1987, respectively, and the Ph.D. degree in electrical engineering from the University of Southern California, Los Angeles, CA, USA, in 1991.

He was an Associate Professor of electrical and computer engineering with the University of Tennessee, Knoxville, TN, USA, and Temple University, Philadelphia, PA, USA. He was the Chair of the Department of Electrical Engineering, Soongsil University, Seoul, and the Graduate Program Director of the Electrical and Computer Engineering Department, Temple University. He is currently a Professor of computer engineering and the Director of strategic planning with Sejong University, Seoul. His research interests include image processing, computer vision, hyperspectral imaging, and intelligent systems.

Dr. Kong was a recipient of the Honorable Mention Paper Award from the American Society of Agricultural and Biological Engineers in 2005 and the Most Cited Paper Award from *Computer Vision and Image Understanding* journal in 2007 and 2008. He was an Associate Editor of IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and the Guest Editor of *Journal of Sensors*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:28:38 UTC from IEEE Xplore. Restrictions apply.